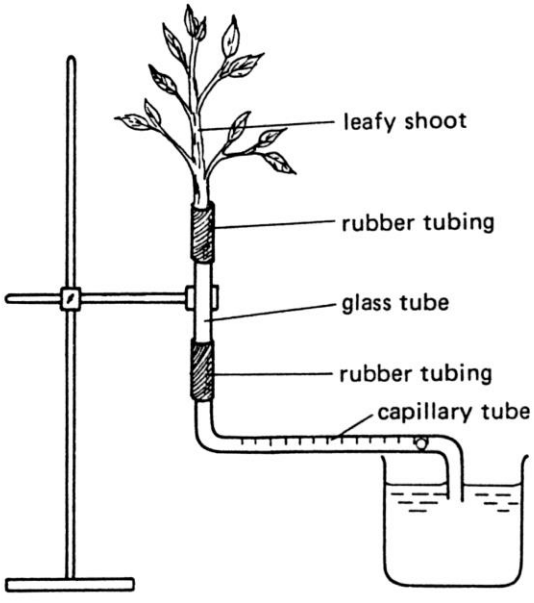
Short questions

|!|BCESB0110000001|!| ★ **SI**

The apparatus below was used to investigate the effect of wind speed on the rate of transpiration of a leafy shoot.



During the investigation, a fan with five different speeds was positioned at a fixed distance from the leafy shoot. The results are shown in the table below.

Wind speed (ms^{-1})	**Rate of water uptake ($mm\ min^{-1}$)**
0	0.30
2	2.54
4	4.92
6	7.08
8	9.45

(a) State ***one*** precaution that should be taken when setting up the apparatus to ensure a continuous stream of water between the capillary tube and the shoot. (1 mark)

_____

_____

(b) Describe and explain the trend shown by the data. (3 marks)

_____

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(c) Explain why the results only give an estimate of the rate of transpiration. (2 marks)

##

(a) Any ***one*** of the following: (1)

- Cut the shoot under water.
- Assemble the whole apparatus under water.
- Ensure tight fit between the shoot and the rubber tubing.
- Seal all joints with vaseline.

(b) The faster the wind speed, the faster the rate of water uptake. (1)

Wind carries away the water vapour around the stomata. The concentration gradient of water vapour between the air spaces and the atmosphere becomes steeper. (1)

Water vapour diffuses out of the leaves more quickly. Hence, the transpiration rate increases. (1)

(c) The apparatus only measures the rate of water uptake by the leafy shoot. (1)

However, some water absorbed is retained in the leafy shoot for photosynthesis, growth and other metabolic activities. Not all the water absorbed is lost by transpiration. (1)

(6 marks)

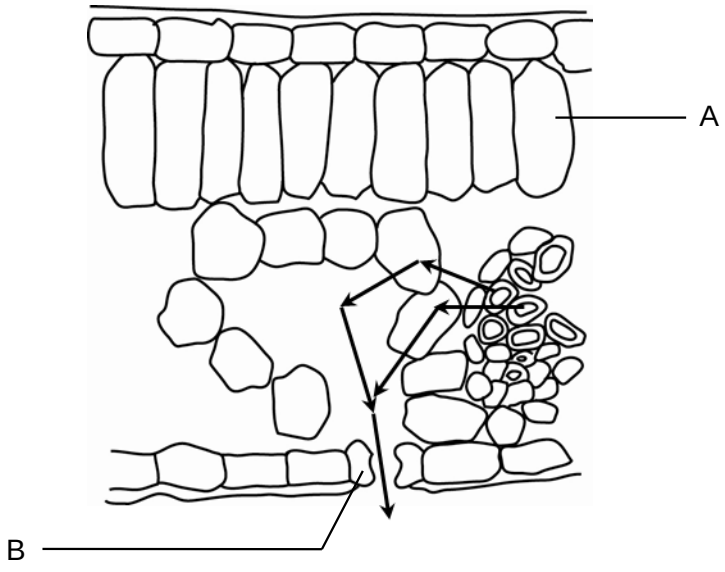
##

	Skill assessed	Mark(s)
(a)	Practical skills	1
(b)	Analysing data	1
	Applying concepts	2
(c)	Applying concepts	2

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|!|BCESC0110000002|!|

The diagram below shows a section of a leaf. The arrows show the pathways of water.



(a) Identify structures A and B. (2 marks)

(b) Describe how water moves in the direction shown by the arrows in the diagram. (4 marks)

##

(a) A: palisade mesophyll cell (1)

B: guard cell (1)

(b) Water vapour diffuses from the air space into the atmosphere through the stoma. (1)

Water evaporates from the surface of the mesophyll cells into the air space. (1)

Water is lost from the mesophyll cells. This lowers the water potential of the mesophyll cells. (1)

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Water is drawn from the xylem vessels by osmosis. (1)

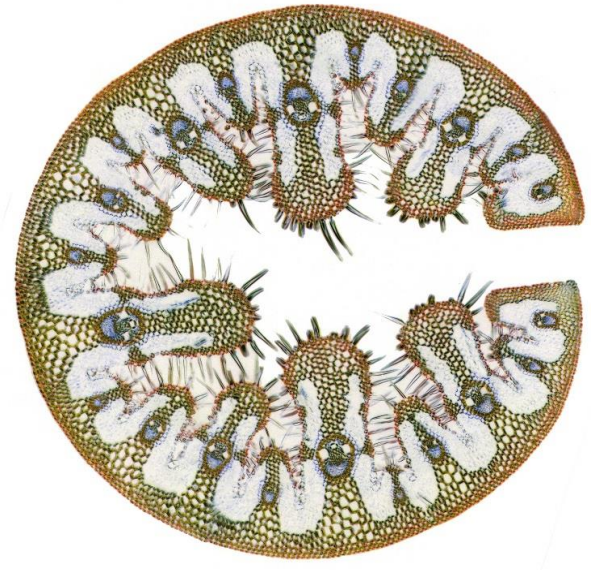
(6 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Understanding basic concepts	4

|!|BCESA0110000003|!| ★★

The photomicrograph below shows the transverse section of a leaf of the marram grass, which is a xerophyte (a plant that lives in dry places where water is scarce).



With reference to the photomicrograph, describe ***three*** features which indicate that this plant is adapted to live in a dry place. For each feature, explain how it helps the plant to survive in such environment. (6 marks)

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##

The leaf rolls up (1)

to trap water vapour inside / to reduce the surface area exposed to environment. (1)

Thick cuticle (1)

to reduce the loss of water by evaporation. (1)

Hairs are present in the upper epidermis (1)

to trap humid air to reduce evaporation. (1)

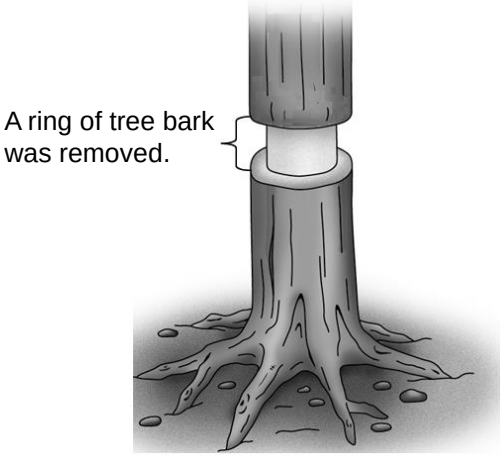
(6 marks)

##

Skill assessed	Mark(s)
Analysing data	3
Applying concepts	3

|!|BCESB0110000004|!| ★

In an investigation, a ring of bark, including the phloem, was removed from the stem of a young tree.



The plant does not die immediately but it dies after several weeks. Explain this.

(6 marks)

##

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Phloem is removed and xylem still exists. (1)

Water and minerals absorbed by the roots can be transported to the aerial parts of the tree. (1)

The leaves can still carry out photosynthesis to produce food. (1)

However, the photosynthetic product (sugars) cannot be transported to the roots. (1)

Once the stored food in the roots is used up, the roots die. (1)

Water and minerals cannot be absorbed. The tree dies. (1)

(6 marks)

##

Skill assessed	Mark(s)
Applying concepts	6

|!|BCESB0110000005|!| ★

The following photographs show water lilies and *Hydrilla*.



Water lilies



Hydrilla

(a) The following table shows the stomatal density of the leaves of four kinds of plants, A to D. Two of the plants are water lily and *Hydrilla*.

Plant	Stomatal density (number of stomata / cm^3)	
	Upper epidermis	Lower epidermis
A	0	0
B	1200	13100
C	9500	0
D	40	50

According to the above table, which plants, A to D, are water lily and *Hydrilla*? (2 marks)

(b) Explain the difference in the number of stomata present in the leaf epidermis between water lily and *Hydrilla*. (4 marks)

##

(a) Water lily: C (1)

Hydrilla: A (1)

(b) Water lily has stomata in the upper epidermis only for gas exchange (1)
as the lower surface is in contact with water and that direct diffusion of gas can take place. (1)
Hydrilla has no stomata at all (1)
as gases can diffuse directly all over its surface. (1)

(6 marks)

##

	Skill assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	4

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|!|BCESB0110000006|!| ★

For each of the term listed in column I, select from column II one phrase that is related to it. Put the letter in the space provided. Some may require more than one answer.

(5 marks)

Column I		***Column II***
Xylem vessels	(a)_____	A. living cells
Transpiration	(b)_____	B. non-living cells
Translocation	(c)_____	C. responsible for the transport of minerals within a plant
Sieve tubes	(d)_____	D. the transport process of organic nutrients within a plant
		E. the loss of water in the form of water vapour from the plant surfaces

##

(a) B, C (1, 1)
(b) E (1)
(c) D (1)
(d) A (1)

(5 marks)

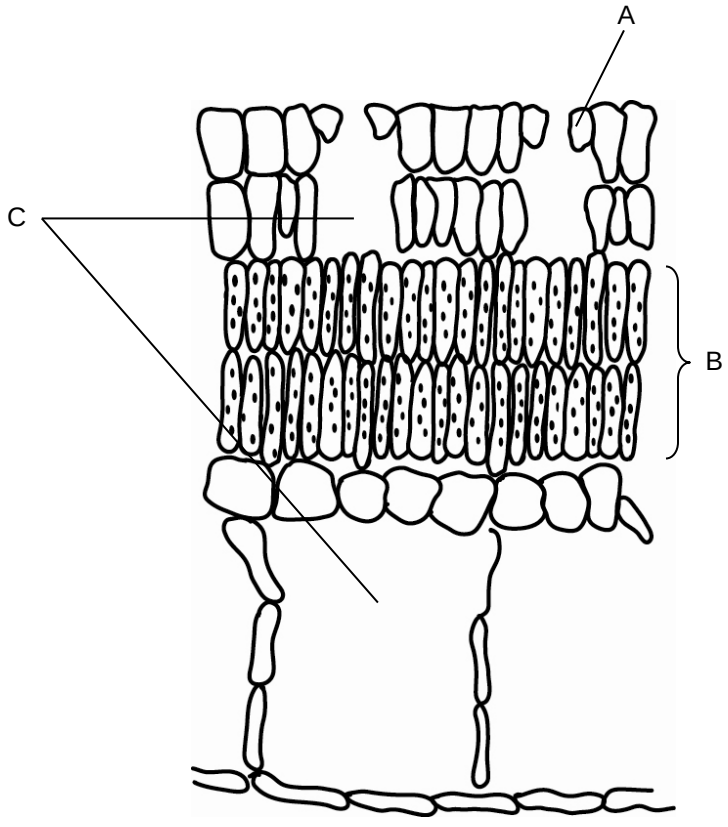
##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

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|!|BCESA0110000007|!| ★★

The following diagram shows the cross section of the leaf of a water lily, with its leaves floating on the surface of pond water.



(a) Identify structures A and B. (2 marks)

(b) State how structures A, B and C can help the plants adapt to live on the surface of pond water. (3 marks)

##

(a) A: guard cell (1)
B: palisade mesophyll (1)

(b) Guard cells (structure A) are found on the upper surface to help gas exchange. (1)

Chloroplasts are concentrated in palisade mesophyll (structure B) to give maximum light absorption for photosynthesis. (1)

Air spaces (structure C) are large to keep the leaf buoyant. / Air spaces are found near the upper surface to facilitate the diffusion of gases from stomata to the mesophyll. (1)

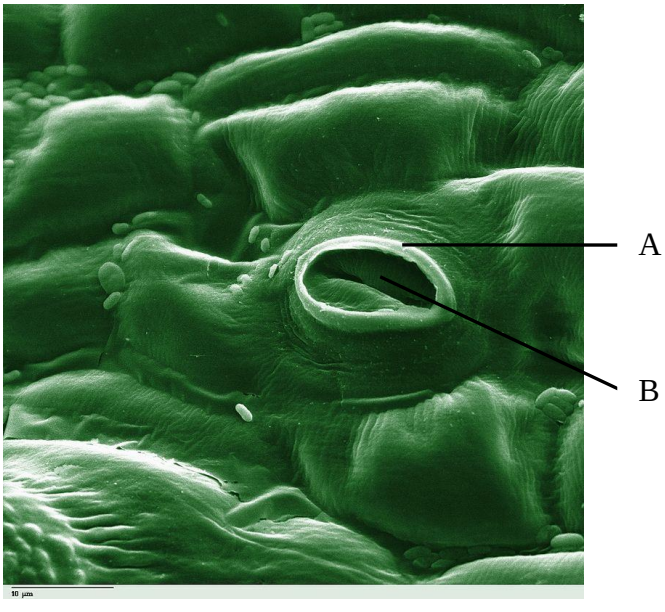
(5 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Understanding basic concepts	3

|!|BCESB0110000008|!| ★

The electron micrograph below shows the epidermis of a leaf.



(a) Name structure A and B. (2 marks)

(b) State the significance of structure A to plants. (1 mark)

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(c) In most terrestrial plants, there is a larger number of structure A at the lower epidermis than the upper epidermis. What is the significance of this? (3 marks)

##

(a) A: guard cell (1)
B: stoma (1)

(b) To allow gases to enter or leave the leaf / gas exchange to take place. (1)

(c) When the plant is under sunlight, the upper surface of the leaves is directly lit by sunlight (1)
and its temperature is higher. (1)
Having fewer structure A / stomata on the upper epidermis helps reduces the transpiration rate of the leaves. (1)

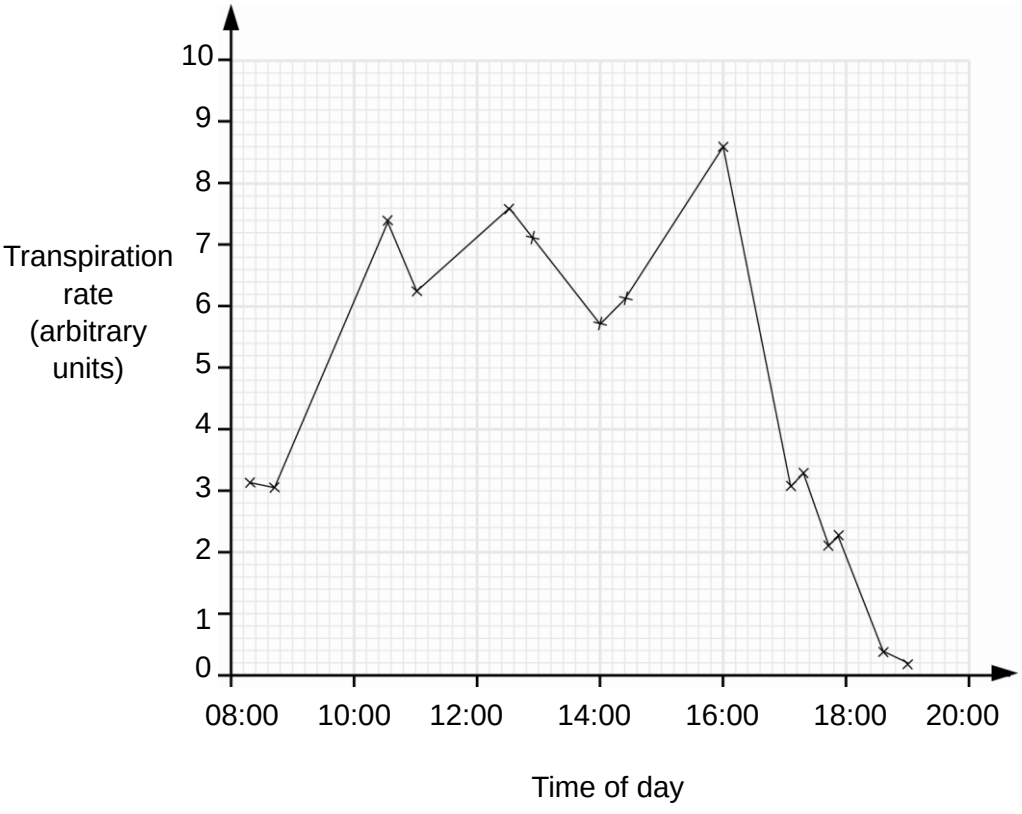
(6 marks)

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	3

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|!|BCESB0110000009|!| ★

The graph below shows the changes of the transpiration rate in a plant at different times of a day.



(a) Through which part of a leaf does most transpiration occur? (1 mark)

_____

(b) According to the graph, the transpiration rate decreases between 16:00 and 19:00. Calculate the average rate of decrease in the transpiration rate over this time. Show your working. (2 marks)

_____

_____

(c) Suggest ***one*** explanation for the decrease in the transpiration rate between 16:00 hours and 19:00 hours. (2 marks)

_____

_____

##

(a) Stoma (1)

(b) (8.6 – 0.2) / 3 (1)

= 2.8 units / hour (1)

(c) Any ***one*** set of the following: (2)

- As the temperature drops from 16:00 to 19:00, (1)
  the evaporation of water gets slower. (1)
- As the light intensity decreases from 16:00 to 19:00, (1)
  less stomata are open and this leads to less water loss by transpiration. (1)

(Or other acceptable answers)

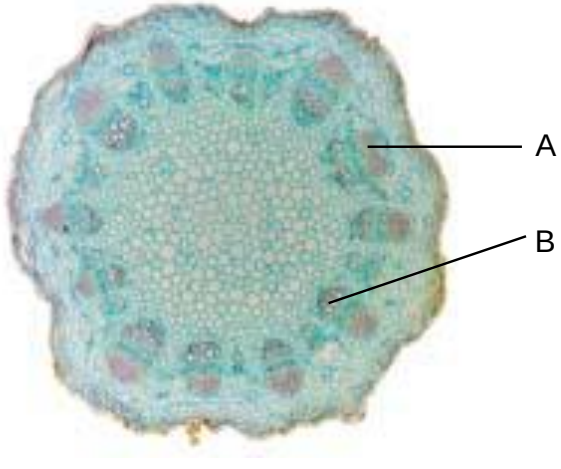
(5 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Mathematical skills	2
(c)	Applying concepts	2

|!|BCESC0110000010|!|

The following photomicrograph shows the cross section of an organ of a plant. A and B are vascular tissues.



(a) To which plant organ does this cross section belong? (1 mark)

(b) Both structures A and B are responsible for transport in the plant. Identify the ***two*** structures and state the substances transported by each of them. (4 marks)

(c) State an advantage for the vascular tissues to locate at the outer regions of the organ. (1 mark)

##

(a) Stem (1)

(b) Structure A: phloem (1)

Structure B: xylem (1)

Substances transported by structure A: organic nutrients (1)

Substances transported by structure B: water and minerals (1)

(c) To provide support to the plant so as to prevent being knocked down by the force generated by nearby animals or wind. (1)

(6 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Analysing data	2
	Understanding basic concepts	2
(c)	Understanding basic concepts	1

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|!|BCESA0110000011|!| ★★ **STSE**

Girdling, which is also known as ring barking, refers to the process of completely removing a ring of bark from a tree. It is commonly used by fruit farmers to yield larger fruits.

(a) Which structure of the tree do the farmers cut out when they remove a ring of bark? (1 mark)

_____

(b) Explain the mechanism by which larger fruits can be yielded by girdling? (3 marks)

_____

_____

_____

_____

(c) Explain if the water transport in the tree is affected when a ring of the bark is removed. (2 marks)

_____

_____

##

(a) They cut out the phloem tissue. (1)

(b) The phloem is responsible for translocation of food (products of photosynthesis). (1)

Removing the phloem prevents food from being transported to the lower parts of the trees. (1)

The nutrients then accumulate in the upper part of the tree (above the ring), causing the fruits in the upper part of the tree to grow larger. (1)

(c) The xylem is responsible for upward transport of water. (1)

As it is located in the inner part of the stem relative to the phloem, its function is not affected. (1)

(6 marks)

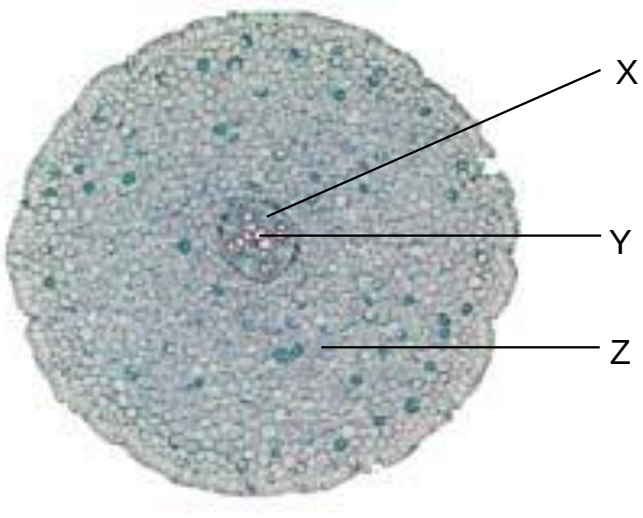
##

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	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	2

|!|BCESC0110000012|!|

The following photomicrograph shows the transverse section of the root of a dicotyledonous plant.



(a) Identify structure Y and state its function. (2 marks)

(b) Suggest ***two*** structural differences between structure X and structure Y. (2 marks)

(c) State ***one*** function of structure Z. (1 mark)

##

(a) Xylem (1)

It transports water and minerals / provides support to the plant. (1)

(b) Any ***two*** of the following: (2)

- Structure Y is made up of dead cells while structure X is made up of living cells. (1)
- The cell walls of cells of structure Y is lignified while those of structure X is not. (1)
- The end walls of cells of structure Y have been broken down while the end walls of cells of structure X still exist and have pores. (1)

(Or other acceptable answers)

(c) Any ***one*** of the following: (1)

- Store carbohydrates (starch) (1)
- Help the root maintain shape (1)
- Help the transport of water in the root (1)

(Or other acceptable answers)

(5 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	1

|!|BCESC0110000013|!|

Fill in the blanks to complete the following paragraph. (6 marks)

The loss of water from the aerial parts of a plant is known as (a)_____. Most water is lost from the leaves. Water evaporates from the surface of (b)_____ _____cells, and the water vapour diffuses out of the air spaces in the leaf through the (c)_____. Due to the loss of water, the (d)_____ _____ of these cells is lowered, and water is eventually drawn from the xylem vessels of the leaf by (e)_____. As a result, (f)_____ _____ is created, which is the main driving force for water transport in the stem.

##

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(a) transpiration (1)

(b) spongy mesophyll (1)

(c) stomata (1)

(d) water potential (1)

(e) osmosis (1)

(f) transpiration pull (1)

(6 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

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